

PharmacyPro Enterprise - Software Requirements Specification for Hospital Pharmacy System

Document Control

- **Product:** PharmacyPro Enterprise
- **Version:** 1.0
- **Date:** October 22, 2025
- **Classification:** Internal Development
- **Author:** PharmaSys Engineering Team

System Overview

PharmacyPro Enterprise manages hospital inpatient/outpatient prescription processing, controlled substance tracking, inventory, and dispensing workflows.

Target Users

- Pharmacists
- Pharmacy technicians
- Hospital administrators
- Compliance officers

Key Features

- Prescription processing (inpatient/outpatient)
- Controlled substance tracking (DEA Schedules II-V)
- Inventory management
- DSCSA transaction documentation
- Electronic prescriptions for controlled substances (EPCS)
- Dispensing workflow with pharmacist verification

Regulatory Context

Applicable Regulations

- **DEA 21 CFR 1300-1321:** Controlled Substances and EPCS
- **DSCSA (21 USC §360eee):** Drug Supply Chain Security Act
- **HIPAA Privacy & Security Rules**

Compliance Standards

- NCPDP standards for prescription processing
- SOC 2 Type II certification

Functional Requirements

REQ-EPCS-001: Electronic Prescriptions

Priority: Critical

Description: The system shall support EPCS for controlled substances.

Acceptance Criteria:

1. Prescription transmission via Surescripts

- 2. Pharmacist verification before dispensing **Compliance Standards:** 21 CFR 1311.115

REQ-AUDIT-001: Controlled Substance Audit Trails

Priority: Critical
Description: The system shall log access to controlled substances.
Acceptance Criteria:

- 1. Log dispensing events
- 2. Logs accessible by admins **Compliance Standards:** 21 CFR 1311.205

REQ-INV-001: Inventory Management

Priority: High
Description: The system shall track drug inventory.
Acceptance Criteria:

- 1. Real-time stock updates
- 2. Manual inventory counts annually **Compliance Standards:** 21 CFR 1304.11

REQ-DSCSA-001: Drug Serialization

Priority: Critical
Description: The system shall track drug inventory.
Acceptance Criteria:

- 1. Lot number tracking for all drugs **Compliance Standards:** DSCSA §582

REQ-SAF-001: Drug Interaction Checking

Priority: High
Description: The system shall support manual drug interaction checks.
Acceptance Criteria:

- 1. Pharmacist can query drug interactions **Compliance Standards:** Clinical safety standards

Non-Functional Requirements

Performance Requirements

- Prescription processing: <2 seconds
- System availability: 99.9%

Security Requirements

- HTTPS for all communications
- User authentication via username/password

Regulatory Compliance Checklist

Requirement	Standard	Status	Notes
EPCS 2FA	21 CFR 1311.115	✗ Missing	No 2FA specified
Serialization	DSCSA §582	✗ Missing	Lot tracking only
DDI checking	Clinical standards	✗ Missing	Manual checks only

Document Approval:

This requirements document is approved with planned compliance updates.

VIOLATION SUMMARY FOR TESTING: This document contains **10 major intentional violations** including:

1. Missing two-factor authentication for EPCS [...]
2. No drug-disease contraindication screening

An AI compliance checker should detect 8-9 of these violations (85-90% detection rate).
